

An In-Depth Examination of SLAM Methods: Challenges, Advancements, and Applications in Complex Scenes for Autonomous Driving

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Abstract—Precise environmental perception and reliable real-time localization are crucial for achieving advanced driver assistance functions. Against this backdrop, Simultaneous Localization and Mapping (SLAM), with its unique advantages, has become one of the indispensable key technologies in the field of autonomous driving (AD). However, as the traffic environment for autonomous vehicles (AVs) continues to become increasingly complex and diverse, such as urban streets, highways, and adverse weather conditions, these challenges pose higher demands on vehicles' localization and mapping capabilities, also bring new opportunities and challenges for further optimization and application of SLAM. In this article, we conduct a comprehensive and in-depth analysis of the current research status and applications of SLAM in complex scenes of AD. First, we explore the challenges faced by AVs in various complex scenes, including the interference of dynamic objects, the precise localization and mapping requirements in large-scale scenes, as well as variable environmental and weather conditions. Subsequently, we delve into the coping strategies and methods of SLAM in these specific scenes. Finally, we compare and summarize the datasets related to SLAM in complex scenes of AD and point out some potential research directions for achieving high-level AD. We hope that this study will track and update the latest progress in SLAM for AVs. To promote the development of the open source community and future academic research, we created a repository <https://github.com/herofly1/CQU-AVL-SLAM> that provides related review papers and methodological resources.

Index Terms—SLAM, autonomous driving, complex scenes, localization.

I. INTRODUCTION

WITH the swift advancement of autonomous driving (AD), the application of Simultaneous Localization and Mapping (SLAM) [1] in complex environments has

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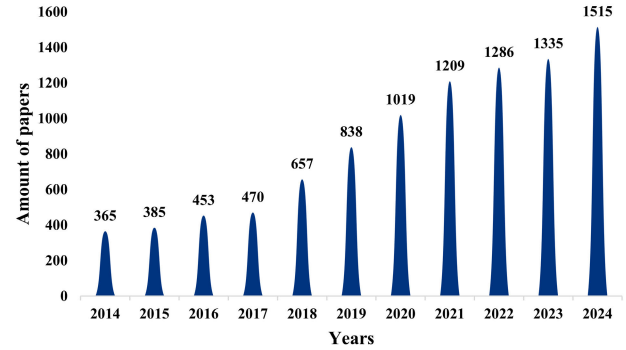


Fig. 1. The growth trend of relevant papers from 2014 to 2024, using SLAM and autonomous driving as search keywords.

emerged as a pivotal research area within the broader field of AD, attracting significant attention and efforts. Fig. 1 shows the growth trend of relevant papers from 2014 to 2024. Autonomous vehicles (AVs) need to achieve precise localization and navigation in various complex environments, and SLAM is the key to addressing this issue. Currently, traditional vehicle localization technologies include Global Navigation Satellite System (GNSS), Inertial Measurement Unit (IMU) localization, and wheel odometer localization. According to the Society of Automotive Engineers (SAE) classification standards for AD, Level 4 and above autonomous vehicles are defined as highly automated vehicles (HAVs). For HAVs, the core challenges and objectives lie in achieving safe and stable navigation amidst the complex real-world scenes. However, the complexity and dynamism of transportation environments, particularly in challenging scenes like urban landscapes with dense edifices [2], confined tunnels [3], and unpredictable outdoor conditions [4], often render traditional localization methodologies inadequate in meeting the stringent accuracy requirements. These limitations pose significant challenges to the safety and reliability of AD systems. However, leveraging its unique advantages, SLAM can fully utilize the environmental information collected by on-board sensors, thereby significantly enhancing localization accuracy and effectively addressing the limitations of traditional localization methods in complex scenes. Therefore, it is particularly important to deeply explore the adaptability of SLAM in complex scenes for AD. The research results will not only promote the

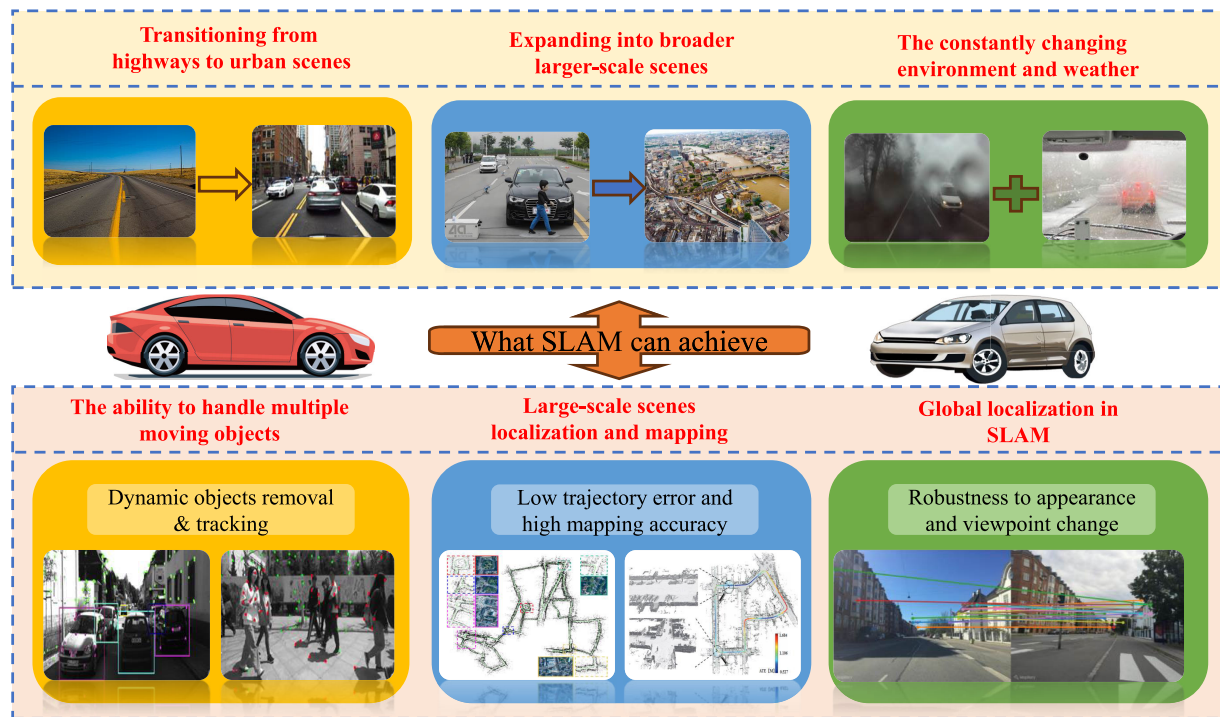


Fig. 2. The challenges faced by AD in complex scenes, as well as the progress and the key role played by SLAM. With a particular emphasis on the advancements made in three core areas: dynamic environment handling, large-scale scenes mapping, and global localization capabilities.

progress of AD, but also have a profound and positive impact on the development of automotive intelligence.

Furthermore, SLAM can be integrated with other traffic-related studies, such as traffic sign interpretation (TSI) [5], traffic knowledge graphs (TKG) [6], and lane detection (LD) [7], which can serve as priors for the SLAM system, to provide a more comprehensive and structured understanding of the traffic environment. This integration enhances the vehicle's ability to interpret complex traffic scenes, respond to dynamic road conditions, and make informed driving decisions, thereby enabling smarter and safer navigation. By adopting this holistic approach, SLAM not only improves the reliability of autonomous systems but also paves the way for advanced driver assistance systems (ADAS) and fully autonomous driving in the future.

Currently, with the continuous advancement and large-scale deployment of high-level autonomous driving, the demand for achieving high-precision localization in various complex scenes is showing a significant growth trend. As shown in Fig. 2, it summarizes the challenges faced by AD in complex scenes, as well as the progress and the key role played by SLAM. We believe that the challenges faced by AVs mainly include the following categories:

Transitioning from highways to urban scenes: The centimeter-level localization accuracy that is easily achieved on highways becomes more challenging to attain in urban environments with towering buildings. Additionally, the large number of dynamic objects in urban environments requires AVs to accurately perceive and flexibly respond to various dynamic changes, such as the real-time movement of other vehicles and pedestrians, to ensure safe and smooth driving.

From small-scale scenes to large-scale scenes: In the past, the testing and validation of AD was mainly focused on small-scale, single scenes such as enclosed industrial parks, university campuses, and carefully designed test sites. These environments typically had low traffic volume and simple road conditions. However, with technological advancements and growing market demand, AVs are gradually entering broader, larger-scale real-world scenes [8]. In these large-scale scenes, the challenges faced by AD systems have also significantly increased, such as reduced localization accuracy, increased cumulative errors, and high computational resource requirements for large-scale map construction.

Variable environments and weather conditions: Due to the complexity of roads and driving environments [9], including various road markings, buildings, natural elements, and texture deficiencies, AVs require significant processing power and adaptability. Additionally, variations in lighting and weather conditions, such as day-night transitions, rain, snow, and other weather phenomena, can potentially interfere with sensors, affecting accuracy and stability.

In recent years, with significant advancements in SLAM, it has provided a range of practical solutions to address the aforementioned complex issues. Specifically, these solutions mainly encompass the following aspects:

- 1) In response to the large number of dynamic objects in the driving environment, SLAM can effectively identify and distinguish between static and dynamic objects. This allows for the precise elimination of dynamic interference or the tracking of dynamic objects, significantly improving localization accuracy and stability.

TABLE I
SUMMARY OF RELEVANT REVIEWS SINCE 2021

Review Title	Ref	Year	Content	Advantages	Disadvantages
A review of visual SLAM methods for autonomous driving vehicles	[10]	2022	Various types of visual SLAM and their applications in AD	Detailed introduction	Lack of long-term and large-scale mapping
Mapping for Autonomous Driving: Opportunities and Challenges	[12]	2021	The challenge of drawing AD maps	Targeted	Not comprehensive enough
Present and Future of SLAM in Extreme Environments: The DARPA SubT Challenge	[14]	2024	Reviewed the progress of SLAM in complex underground environments	Targeted	Lack of analysis on difficult land scenes
Autonomous driving system: A comprehensive survey	[15]	2024	Comprehensive progress in various fields of AVs	Comprehensive	A comprehensive review with inadequate specificity
Neural Network-Based Recent Research Developments in SLAM for Autonomous Ground Vehicles: A Review	[13]	2023	Recent Developments in Neural Network Techniques for SLAM-Based AVs	Targeted	Not comprehensive enough
A comprehensive overview of dynamic visual SLAM and deep learning: concepts, methods and challenges	[16]	2022	Processing Dynamic VSLAM problems based on deep learning	Detailed introduction	Not comprehensive enough
Simultaneous Localization and Mapping (SLAM) for Autonomous Driving: Concept and Analysis	[11]	2023	The application and challenging issues of SLAM in AD	Comprehensive	Insufficient in-depth analysis
A Survey on Global LiDAR Localization: Challenges, Advances and Open Problems	[17]	2024	Detailed introduction of global LiDAR localization	Targeted	Lack of correlation analysis with real-time localization

- 2) Utilizing SLAM, vehicles can construct high-precision, large-scale maps covering tens of kilometers, while crowd-sourced mapping techniques further enable rapid map updates. Additionally, with the continuous advancement of scene reconstruction technology, it will provide more precise and abundant data support for simulation testing of AVs.
- 3) By leveraging the global localization technology within SLAM, vehicles can correct cumulative errors during long-distance localization in variable environments and weather conditions, thus achieving long-term anti-interference and high-precision localization capabilities.

Based on the aforementioned three points, we will conduct an in-depth analysis of the specific solutions for each aspect in Section II.

Over the past few years, with the development of AD and the widespread application of SLAM in this field, it is necessary to conduct a comprehensive review of the challenges and progress of SLAM in complex scenes. However, it is worth noting that existing review papers either lack sufficient coverage or lack in-depth analysis of complex scenes in AD. As shown in Table I, we have summarized relevant review papers since 2021. Reference [10] provided a basic overview of the application of Visual SLAM (VSLAM) in AD, [11] comprehensively summarized the importance of SLAM to AD and its current progress, while [12] and [13] analyzed the applications of map construction and Neural Networks to SLAM.

Unfortunately, these review papers do not provide an in-depth analysis of the complex scenes in AD, nor do they analyze the challenges associated with the long-term usability of the maps they create. Additionally, [14] focuses more on an in-depth analysis of the challenges and progress of SLAM in extreme underground environments, with less emphasis on difficult terrestrial scenes. Reference [17] is the first

review from the perspective of lidar global localization, but it lacks an analysis of the relevance to real-time localization. Reference [16] discusses dynamic SLAM more from the perspective of deep learning, but its coverage is insufficient. Reference [15] discusses AVs from the level of the entire AD system, rather than limiting itself solely to the research area of SLAM.

Given the gaps in existing literature, it is necessary to conduct an in-depth summary of the challenges and progress of SLAM in complex scenes for AD. This will provide essential insights for researchers in the field of SLAM research in dynamic environments, large-scale scene mapping, severe weather conditions, as well as a diverse array of other complex scenes that are intimately related to AVs navigation. Notably, in recent years, SLAM research has also made significant progress, such as the superior performance of SLAM in scene reconstruction based on Neural Radiance Fields (NeRFs) [18] and 3D Gaussian Splatting (3DGS) [19], and the application of large language models (LLMs) significantly improves the understanding of the environment. Therefore, it is also necessary to summarize these latest developments in a timely manner.

Specifically, our main contributions are as follows:

- We conduct a comprehensive analysis of current SLAM methods, focusing particularly on complex scenes in AD. The discussion is conducted from three main perspectives: handling of dynamic environments, construction of large-scale scene maps, and long-term anti-interference capabilities of SLAM. To our knowledge, this is the first review to discuss SLAM from the perspective of its application in complex scenes for AD.
- We compile datasets related to complex scenes in AD, which will be instrumental for future researchers to continue their investigations and improve existing SLAM methods.

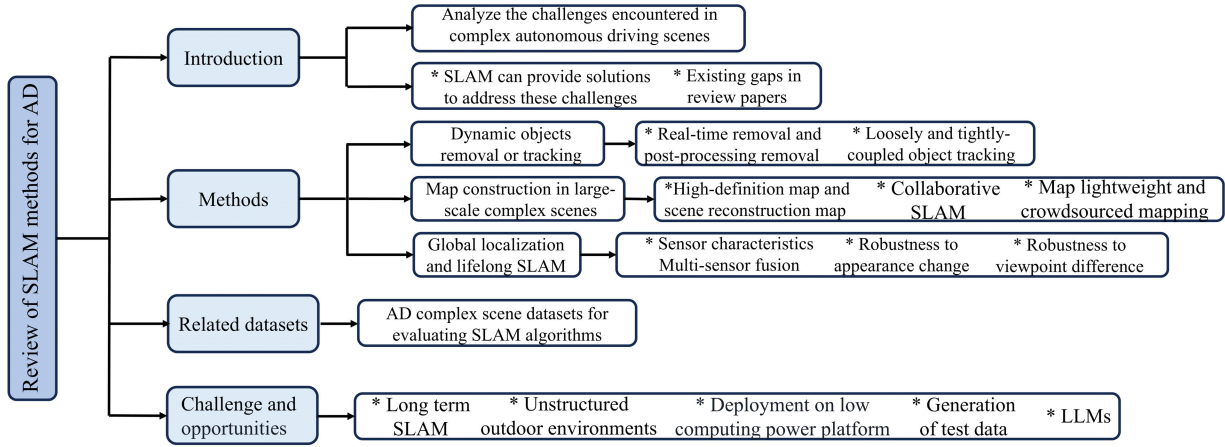


Fig. 3. The organizational structure of this paper's content.

- We present current challenges and proposed further research directions, which will assist researchers in conducting in-depth studies on SLAM issues in complex scenes for AD.

The structure of this paper is shown by Fig. 3. The paper is organized as follows. Begins by examining autonomous driving challenges and SLAM's role in Section I. Section II analyzes various SLAM methods. Section III summarizes relevant datasets. Finally, Sections IV and V discuss future research trends and conclusions, respectively.

II. METHODS

In the field of AD, SLAM is crucial for enabling vehicles to navigate autonomously in unknown environments [20]. Compared to traditional applications, SLAM in complex scenes of AD faces a series of unique challenges and specific requirements.

- 1) AVs are often required to navigate in complex and dynamic environments, such as urban roads, highways, tunnels, and bridges. These environments contain a large number of obstacles, pedestrians, vehicles, and other dynamic elements, which may not have been considered in traditional SLAM applications.
- 2) AD places stringent demands on the accuracy of localization and map construction. Vehicles need to have a precise understanding of their location on the road in order to perform accurate navigation, obstacle avoidance, and path planning.
- 3) To meet the demands of AD, SLAM systems typically require a more complex configuration of sensors, including lidar, high-definition cameras, IMU, and others. These sensors provide rich environmental information, helping the SLAM system to perceive and construct an environmental model with greater accuracy.
- 4) The data processing capability is more demanding, SLAM in complex scenes of AD needs to process a large amount of sensor data and make decisions in a short period of time. This poses a challenge to the real-time performance of the system.

To address the aforementioned challenges, this section will delve deeply into the most suitable and effective SLAM methods for the field of AD, aiming to improve the system's accuracy, real-time performance, and robustness to meet the stringent requirements of AD.

To comprehensively evaluate the performance of different SLAM methods, we have adopted a series of standardized metrics. Firstly, we utilize FPS (Frames Per Second) to measure the real-time processing capability of SLAM methods, which refers to the number of frames processed per second. This metric directly reflects the real-time performance of the methods, with higher frame rates indicating better real-time capabilities. Secondly, to assess the accuracy of the methods, we employ the RMSE (Root Mean Square Error) of ATE (Absolute Trajectory Error) as a criterion. This metric is applicable for evaluating performance in areas such as dynamic object filtering, map construction, and loop closure detection, with smaller ATE values indicating superior performance in these aspects. Additionally, we evaluate the robustness of the methods across three dimensions: dynamic object interference (D), lighting variation interference (L), and scene change interference (S). Following the practice in reference [21], we adopt a subjective evaluation using "Tick" or "Cross", where "Tick" indicates good robustness of the SLAM method under the aforementioned interference conditions, and "Cross" indicates poor robustness. When assessing system operational costs, we primarily focus on the hardware costs required by the system. Similarly, we use the subjective evaluation method of "Tick" or "Cross", where "Tick" represents satisfaction with the hardware costs, and "Cross" represents dissatisfaction, with corresponding reasons provided.

It should be noted that due to the varied types of test datasets used for experimental verification in this paper, the evaluation metrics cannot provide an entirely fair comparison of the strengths and weaknesses of each method.

A. Dynamic Objects Removal or Tracking

In AD applications, SLAM algorithms face significant challenges due to the dynamic nature of real-world environments

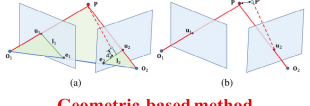
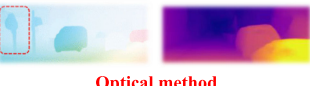
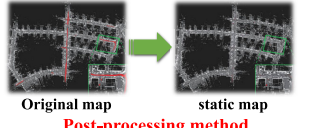
Different methods	Advantages	Limitations
 Geometric-based method	(1) Simple formulation & implementation for efficiency (2) Easy deployment (3) Excellence in structured, geometric environments (4) Accurate, stable SLAM in controlled settings	(1) Relies on quality & density of environmental features (2) Struggles in high-dynamic, crowded environments (3) Limited contextual understanding (4) Depends on initial conditions, may degrade over time
 Objects detection and segmentation	(1) Enhances localization accuracy & robustness (2) Improved map consistency (3) Reduced false positives risk (4) Easy integration with prebuilt maps	(1) Potential important feature loss (2) Computational complexity (3) Depends on accurate segmentation (4) Difficulty with semidynamic objects
 Optical method	(1) High sensitivity to minute movements (2) Capability to track non-rigid surfaces	(1) Computational complexity & real-time challenges (2) Susceptible to illumination & noise (3) Motion model limitations (4) Depends on scene prior knowledge
 Post-processing method	(1) Enhanced map quality (2) Robustness improvement (3) Optimized resource utilization (4) Support for multi-sensor fusion	(1) Computational resource consumption (2) Algorithm complexity (3) Dependence on sensor accuracy
 Multi object tracking	(1) Comprehensive environmental understanding (2) Enhanced robustness in dynamic settings (3) Uses dynamic objects as landmarks (4) Anticipates future positions (5) Improves interaction with surroundings	(1) Computational intensity (2) Data association complexity (3) Potential for tracking errors (4) Vulnerability to rapid movements (5) Sensor dependence increase (6) Long-term consistency challenge

Fig. 4. Summarize the different processing methods for SLAM in dynamic environments.

[22], [23]. Dynamic objects such as pedestrians and vehicles can cause inconsistencies in the generated map, leading to drift and inaccuracies over time. Moreover, moving objects can interfere with the tracking of key features, complicating the maintenance of a stable and accurate representation of the environment. The presence of dynamic elements also increases the computational load, as the system must process and filter out moving objects in real-time, which is critical for ensuring that the SLAM system remains robust and reliable for safe navigation. To address these issues, two primary approaches have been proposed [24]: one identifies and removes dynamic features as outliers to minimize interference and maintain stability, while the other actively tracks and utilizes dynamic features to enhance the system's adaptability and understanding of the environment. Different SLAM methods vary in their effectiveness in handling dynamic scenes, as illustrated in Fig. 4, and selecting or developing solutions that effectively tackle these challenges is essential for improving the performance of AD systems.

1) *Detection & Removal*: In the SLAM system, if the feature points extracted from dynamic objects in the previous frame are projected onto the current frame, due to the movement of these dynamic objects between the two frames, the matched points found will inevitably be incorrect. This incorrect matching will significantly reduce the accuracy of pose estimation, thus requiring the removal of dynamic features.

a) *Geometric-based method*: In the early stage, researchers identified dynamic objects through geometric methods [25], [46] and employed an improved Random Sample Consensus (RANSAC) algorithm [47] to eliminate

dynamic feature points. CoSLAM [26] identified dynamic map points by analyzing their triangulation consistency. Li and Lee [25] employed a static weighting strategy to reduce the impact of dynamic objects on pose estimation. This strategy involves assigning a likelihood of static nature to each feature within keyframes, which is then integrated into the intensity-assisted iterative closest point algorithm for camera pose estimation.

In addition to point features, line, and planar features find applications in geometric-based dynamic VSLAM methods. Zhang et al. [27] introduced DynPL-SVO, a stereo visual SLAM approach that leverages both point and line features. By utilizing motion models and dynamic grids, it detects outliers and mitigates the influence of dynamic objects by excluding features located within these grids. Fan et al. [28] presented a SLAM method for dynamic scenes that focuses on planar features. Utilizing a deep learning network, this approach detects static planar features and integrates planar homography constraints into the reprojection error to expedite the optimization process.

b) *Segmentation and detection-based method*: Currently, methods based on semantic segmentation and object detection are the mainstream approaches for dynamic VSLAM. Semantic segmentation methods boast high accuracy but require significant computational resources, while object detection methods are faster but with slightly lower precision. Reference [29] combined semantic segmentation algorithms with geometric methods to identify and eliminate dynamic objects in images. DS-SLAM [30], an improvement on ORB-SLAM2 [48], first utilizes SegNet [49] for semantic segmentation, followed by a motion consistency check to eliminate feature

points in dynamic regions. Based on this, [50] proposed a method that combines semantic segmentation information and spatial motion information. Similar to DS-SLAM, DynaSLAM Reference [29] utilizes Mask R-CNN [51] and geometric methods to detect potentially moving and moving objects, respectively. It estimates the pose only using ORB feature points located in static regions and the edges of non-dynamic object masks.

Some methods focus on enhancing the real-time performance of the system. RDS-SLAM [52], for instance, employs a Bayesian filter to identify and eliminate dynamic objects effectively. It also introduces a novel semantic segmentation thread operation mode to mitigate the impact on SLAM's real-time performance. To further reduce the computational burden, approaches like [22] and [53] incorporate object detection. However, a common challenge with these methods is that object detection bounding boxes often include large background regions, potentially causing the removal of static feature points and reducing system stability. Dynamic-VINS [31] addresses this issue by combining object detection with depth information for dynamic feature recognition. This approach achieves performance comparable to semantic segmentation while leveraging IMU for motion prediction, feature tracking, and motion consistency checks. Another notable method, RLD-SLAM [54], integrates object detection with Bayesian filtering to maintain high accuracy while swiftly acquiring static feature points.

With the advancements in deep learning, the speed and accuracy of detecting dynamic objects in point clouds have significantly improved. Reference [32] proposes the use of the 3D-MinNet network for real-time dynamic filtering, followed by the LOAM [55] algorithm to process the lidar point cloud after filtering out dynamic objects. This marks the first general-purpose dynamic object perception Lidar SLAM (LSLAM) that utilizes unsupervised learning.

In recent years, researchers have continuously optimized learning-based dynamic point cloud filtering algorithms to ensure real-time performance on autonomous vehicles (AVs). The method presented in [56] leverages continuous range images from lidar sensors as an intermediate representation, combined with convolutional neural networks, to achieve a processing speed faster than the sensor's frame rate. Meanwhile, [33] uses a sparse 4D CNN to jointly extract spatial-temporal features based on scan-to-map differences and relative timestamps between points. The M-detector [34], designed based on the principle of occlusion, exhibits excellent accuracy, computational efficiency, detection latency, and generalization ability. These advancements demonstrate the ongoing efforts to enhance the real-time performance of SLAM systems in dynamic environments.

c) Optical method: In recent years, some researchers have proposed dynamic VSLAM methods based on optical flow or scene flow segmentation. The sensitivity of optical flow to subtle movements and its ability to track non-rigid surfaces in motion make it suitable for estimating pixel movements between images. YO-VIO [57] utilizes optical flow to identify moving objects and performs localization after removing feature points. Reference [35] first employs a motion

segmentation network to extract dynamic optical flow, setting the pixel values in dynamic optical flow regions to 0, and then jointly optimizes the motion segmentation results and pose estimation. RSO-SLAM [36] integrates an optical flow attenuation propagation strategy to facilitate detailed motion probability calculations and inter-frame propagation within each recognized region. It is worth noting that the stability of optical flow-based methods still needs to be enhanced, and they have high computational resource requirements for the system.

d) Post-processing method: Due to the high measurement accuracy and stability of lidar sensors, LSLAM is less affected by dynamic objects during the point cloud registration phase. However, most challenges arise during the point cloud map construction phase. For autonomous vehicles (AVs), low-quality maps resulting from unfiltered dynamic objects can lead to errors in path planning and obstacle avoidance, compromising safety and efficiency. To enhance the accuracy and completeness of dynamic point cloud filtering, a post-processing approach is employed. This method leverages multi-frame information to achieve more thorough removal of dynamic objects without requiring significant real-time processing. As a result, it improves map quality and enhances the reliability of subsequent tasks. Currently, there are three primary methods for post-processing: Segmentation-based, Ray-casting based, and Visibility-based.

Segmentation-based methods typically rely on clustering [58], [59], such as the Viewpoint Feature Histogram (VFH) proposed in [59], which utilizes feature distribution histograms from a determined viewpoint. Reference [58] suggests that points with large matching errors during the registration of adjacent frames are likely to be dynamic points. These points are used as seeds for region growing, and the resulting clusters are identified as dynamic clusters. Additionally, semantic segmentation method has significantly improved the performance of dynamic element detection [37]. Given correctly segmented points with dynamic labels, maps can be directly constructed by excluding the predicted dynamic points.

Ray-casting based methods require the integration of grids, which can be either traditional occupancy grids or octree grids. Ray-casting based methods are computationally expensive, so some approaches focus on improving computational speed. For instance, [38] proposes a method that accelerates grid traversal and reduces the consumption of computational resources.

Visibility-based methods are based on a fundamental assumption [39]: if a previously acquired point in the map is re-queried during subsequent scans, it is considered dynamic. However, this assumption has limitations when dealing with large lidar incidence angles and occlusions by dynamic objects. In contrast, the method proposed in [40] overcomes the limitations of both ray-casting and visibility-based approaches and achieves good results in dynamic filtering.

2) Detection and Tracking: Multi-Object Tracking SLAM (MOT-SLAM) has emerged as a research hotspot in recent years. Researchers in this field have acknowledged the limitations of directly discarding dynamic features, particularly in scenes where dynamic objects occupy a significant

proportion. Simply discarding dynamic features can lead to an insufficient number of feature points for pose estimation, thereby affecting the stability and positioning accuracy of the SLAM system. Therefore, effectively utilizing dynamic features while avoiding their interference in pose estimation has become a crucial research direction in current MOT-SLAM. Reference [60] first constructed a mathematical framework integrating filtering-based SLAM with moving object tracking, effectively meeting navigation and safety demands in AD. Currently, there are two main approaches for integrating object tracking tasks into SLAM or odometry systems. The first approach [41] involves loosely coupling the odometry and dynamic object tracking optimizations, where the independent tracking results are used to further refine the odometry. The second approach [42] combines the odometry system and object tracking task into a tightly coupled state estimation problem for joint optimization.

a) *Loosely-coupled object tracking*: The loosely-coupled approach consists of two independent systems, one for odometry localization and the other for multi-object tracking. Reference [61] proposes using one system to estimate the camera trajectory and another to track moving pedestrians, where the absolute scale of the trajectory is recovered based on the given height of the tracked individuals. ClusterSLAM [62] designs a practical backend for stereo VSLAM, which performs hierarchical clustering of dynamic objects in the scene and utilizes a decoupling factor to correct the trajectories of dynamic objects. ClusterVO [41], building upon ClusterSLAM, introduces a multi-level probabilistic association mechanism and a heterogeneous Conditional Random Field (CRF) clustering method. This method combines semantic, spatial, and motion information to jointly infer the clustering segmentation of each frame online. Simultaneously, the asynchronous state estimation method employed reduces the frequent adjustments to the factor graph architecture, effectively minimizing the consumption of computational resources.

b) *Tightly-coupled object tracking*: The tightly coupled approach aims to jointly optimize and solve the multi-object tracking and odometry systems, yielding more accurate state estimation results compared to the loosely coupled method. Reference [63] first proposed integrating the motion information of dynamic objects into the SLAM system. Reference [64] suggested representing objects using 3D bounding boxes estimated from 2D bounding boxes and employing 2D visual tracking algorithms to track dynamic objects. Reference [65] first performed depth estimation, followed by instance-level object segmentation and optical flow estimation. Then, it estimated the camera pose and dynamic object motion separately, finally optimizing through Bundle Adjustment (BA). Twist-SLAM [43] first conducted panoramic segmentation of images, then created 3D semantic point clusters by fusing multiple 2D observations. By defining constraints between clusters modeled by mechanical joints, it jointly estimated the poses of dynamic objects and the camera, improving optimization accuracy. Reference [66] employs neural network-based monocular depth estimation to generate the 3D position of dynamic objects. AirDOS [42] is the first dynamic object-aware SLAM system that introduces rigid and motion constraints into the

model for articulated objects, jointly estimating camera poses, object motions, and object structures within a factor graph.

It is noteworthy that most existing tightly coupled multi-object tracking VSLAM systems perform object detection and tracking in 2D images, which can lead to inaccurate 3D object representations. Additionally, texture loss and low-light environments can also interfere with the system. Recently, some researchers have proposed Lidar-based methods. For instance, Tian et al. presented DL_SLOT [44], which is particularly suitable for dynamic road scenes. This method integrates the state estimation of dynamic objects and the ego-vehicle into a unified optimization framework, demonstrating that the pose errors of dynamic objects primarily stem from the rotational errors of the ego-vehicle. The recently proposed LIO-SEGMOT [45] introduces a hierarchical criterion for coupling and tracking to overcome the uncertainties in object detection, enhancing the system's robustness. The recently proposed IMM-SLAMMOT [67] integrates Interacting Multiple Model (IMM) into the system and proves to be more suitable for complex realistic dynamic scenes.

The performance evaluation of some SLAM methods in dynamic scenes is summarized in Table II.

B. Map Construction in Large-Scale Complex Scenes

As the level of AD increases, the reliance on maps continues to intensify, with AVs requiring precise maps to identify crucial information such as roads, traffic signals, and obstacles. SLAM generates high-precision environmental maps by collecting environmental data and utilizing relevant algorithms to transform the data into an understanding of the environment. Furthermore, with the application of SLAM in large-scale scene reconstruction, the constructed environmental maps can be utilized for functional testing, closed-loop simulation, and other purposes in AVs. The map constructed by SLAM is shown in Fig. 5.

a) *High-definition map construction*: High-definition (HD) maps are specifically designed for AD, comprising instance-level vectorized representations of map elements such as crosswalks, lane dividers, and road boundaries. These maps also encapsulate extensive semantic information on road topology and traffic rules, which is crucial for the navigation of AVs [68]. Table III outlines the varying map requirements across different levels of AD. Due to the stringent precision requirements, the mapping process necessitates the fusion of sensor data from Cameras, LiDARs, GNSS, and Inertial Navigation Systems (INS). However, this fusion process demands substantial computational power, which is currently infeasible for in-vehicle systems. Consequently, traditional HD map construction based on SLAM (Simultaneous Localization and Mapping) has adopted an offline approach, involving multiple acquisitions of road network information, followed by processing the collected sensory data alongside location information to enhance mapping accuracy. [69] constructed the topology of intersections by fitting straight lines between entry and exit points, while employing cubic spline curves for detailed refinement to further refine the geometric structure of virtual lanes. Reference [70] proposed the use of segmented

TABLE II

ELIMINATION AND TRACKING OF DYNAMIC OBJECTS METHODS REGARDING SLAM CRITERIA. “✓”: CRITERION SATISFIED. “×”: CRITERION NOT SATISFIED. “—”: NOT MEASURED. D: ANTI INTERFERENCE ABILITY FOR DYNAMIC ELEMENTS, L: ANTI INTERFERENCE ABILITY FOR LIGHT CHANGE, S: ANTI INTERFERENCE ABILITY FOR SCENE VARIATION

Method	FPS	ATE(m)	Robustness			System cost		
[25]	18	≤0.470 (indoor)	D:✓	L:×	S:—	Time cost:✓	Hardware cost:✓	
[26]	26	≤0.374 (indoor)	D:✓	L:—	S:—	Time cost:✓	Hardware cost:✓	
[27]	—	≤2.631 (outdoor)	D:✓	L:—	S:✓	Time cost:✓	Hardware cost:✓	
[28]	13	≤0.613 (indoor)	D:✓	L:—	S:—	Time cost:✓	Hardware cost:✓	
[29]	—	≤1.600 (outdoor)	D:✓	L:×	S:—	Time cost:×	Hardware cost:×	(Large memory usage)
[30]	14	≤0.827 (indoor)	D:✓	L:—	S:×	Time cost:✓	Hardware cost:✓	
[31]	22	≤0.062 (outdoor)	D:✓	L:—	S:✓	Time cost:✓	Hardware cost:✓	
[32]	25	≤0.810 (outdoor)	D:✓	L:—	S:—	Time cost:—	Hardware cost:✓	
[33]	—	≤0.631 (outdoor)	D:✓	L:—	S:✓	Time cost:✓	Hardware cost:✓	
[34]	21	≤0.903 (outdoor)	D:✓	L:—	S:✓	Time cost:✓	Hardware cost:✓	
[35]	25	≤0.637 (outdoor)	D:✓	L:—	S:✓	Time cost:✓	Hardware cost:×	(Need GPU acceleration)
[36]	22	≤0.269 (outdoor)	D:✓	L:—	S:—	Time cost:✓	Hardware cost:✓	
[37]	27	≤0.814 (outdoor)	D:✓	L:—	S:✓	Time cost:✓	Hardware cost:✓	
[38]	offline processing	— (outdoor)	D:✓	L:—	S:—	Time cost:—	Hardware cost:✓	
[39]	offline processing	— (outdoor)	D:✓	L:✓	S:✓	Time cost:×	Hardware cost:✓	
[40]	offline processing	— (outdoor)	D:✓	L:—	S:×	Time cost:✓	Hardware cost:✓	
[41]	8	≤3.180 (outdoor)	D:✓	L:—	S:×	Time cost:×	Hardware cost:✓	
[42]	19	≤2.710 (outdoor)	D:✓	L:×	S:—	Time cost:✓	Hardware cost:✓	
[43]	15	≤0.260 (outdoor)	D:✓	L:—	S:✓	Time cost:✓	Hardware cost:—	
[44]	19	≤0.418 (outdoor)	D:✓	L:—	S:—	Time cost:✓	Hardware cost:✓	
[45]	—	≤0.910 (outdoor)	D:✓	L:—	S:✓	Time cost:✓	Hardware cost:✓	

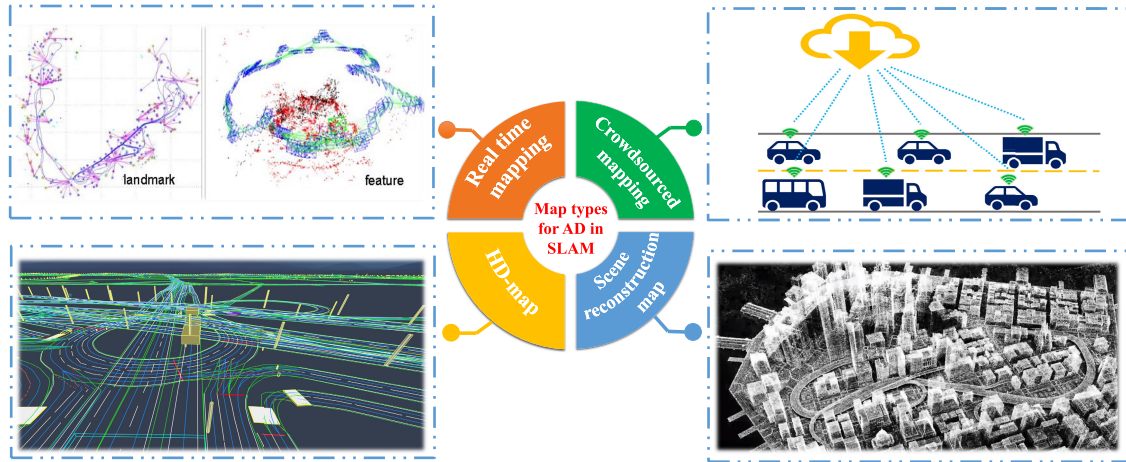


Fig. 5. Map types built by SLAM for AD include sparse feature point maps constructed during real-time localization, high-definition maps with vectorized features for navigation, crowdsourced mapping that reduces hardware consumption, and dense scene reconstruction maps.

cubic spline curves and control points to address the geometric shape of lanes.

In recent years, due to the high maintenance costs associated with offline maps, an increasing number of researchers have begun to focus on the construction of online high-precision maps. AVs in motion autonomously construct environmental maps using onboard sensors, minimizing human intervention. Reference [71] proposed a VSLAM system employing biomimetic eyes that can actively search for texture-rich regions in outdoor, weakly textured environments, enhancing the system's robustness for operation in large-scale outdoor scenes. Additionally, a SLAM system integrating cameras

and LiDARs has been proposed [72], effectively improving mapping quality in unstructured environments. Currently, the integration of SLAM and deep learning techniques is also widely applied to online inference of road structures. Reference [73] presented a novel learnable representation of local road networks that combines the advantages of directed graphs and B-spline curve centerlines. This representation is used to train a transformer-based neural network architecture capable of predicting the complete lane graph structure from a single onboard image. With the development of bird's eye view (BEV) representation learning, some researchers have generated rasterized maps through BEV semantic segmentation [74].

TABLE III
REQUIREMENTS OF HD MAP FOR DIFFERENT-LEVELS AVs

SAE Levels	Control	Forms of map	Map	Accuracy of map	Use of map	Necessity
L1	Human & System	Static map	Navigation digital map	Meter-level to Sub-meter level	In-vehicle navigation	Options
L2	Human & System	Static map	Navigation digital map & ADAS map	Meter-level to Sub-meter level	Auxiliary driving	Options
L3	System	Static map + Dynamic traffic information	ADAS Map + HD Map	Sub-meter to Centimeter-level	Auxiliary driving	Options
L4	System	Static map + Dynamic traffic information	ADAS Map + HD Map	Sub-meter to Centimeter-level	Autonomous Driving	Mandatory option
L5	System	Static map + Dynamic map + Dynamic traffic information + Analyze data intelligently in real time + events	HD Map	Centimeter-level	Autonomous Driving	Mandatory option

HDMaNet [75] predicts the surrounding environmental map of AVs through neural networks and obtains the vectorized topological relationships of roads after post-processing. To further reduce post-processing operations, VectorMapNet [76] and MAPTRV2 [77] have been developed, enabling vectorized mapping results to be directly output in an end-to-end manner through the network. MapLite 2.0 [78] designed a perception and reasoning system that estimates high-precision map model parameters while achieving rapid detection of environmental changes and real-time map updates. The recently proposed MapQR [79] focuses on enhancing querying capabilities for building online vectorized maps using instance queries rather than point queries.

b) Scene reconstruction map: AVs must accurately perceive and interpret their surroundings to ensure safe driving. This necessitates the ability to reconstruct driving scenarios in real-time and efficiently identify dynamic objects such as pedestrians and other vehicles. The development of AVs is also a data-driven process that requires extensive and in-depth testing across diverse driving scenarios. To this end, practitioners must collect vast amounts of driving scenario data to support the continuous optimization and performance enhancement of autonomous driving systems. Currently, advancements in scene reconstruction technologies will help address these issues.

Early approaches focused on light field and basic scene reconstruction methods. The work in reference [80] presented the first dense SLAM system capable of reconstructing non-rigidly deforming scenes in real-time, while [81] learned and maintained 3D models for each dynamic object, improving model outcomes through temporal fusion. Reference [82] tracked dynamic objects as rigid bodies, enhancing mapping quality. However, these methods, heavily reliant on dense sampling and constraints from structured environments, faced challenges when dealing with large-scale complex outdoor scenes and diverse lighting conditions. This limitation undoubtedly constrained their wide applicability and practical effectiveness in the AD domain. R3LIVE [83] proposed by Lin and Zhang, achieved precise and dense reconstruction of the surrounding environment while enhancing the system's robustness in degraded lidar conditions. Liu et al. introduced HBA [84], a globally consistent and efficient mapping method

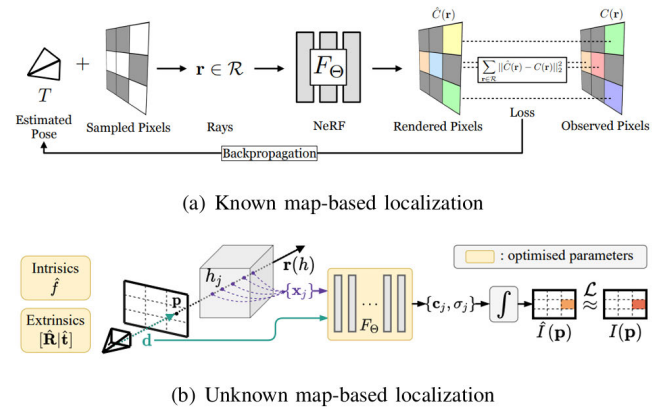


Fig. 6. (a) Optimizes attitude estimation and reconstruction quality using pre-trained NeRFs; [86] and (b) jointly optimizes the camera pose and model properties. [87].

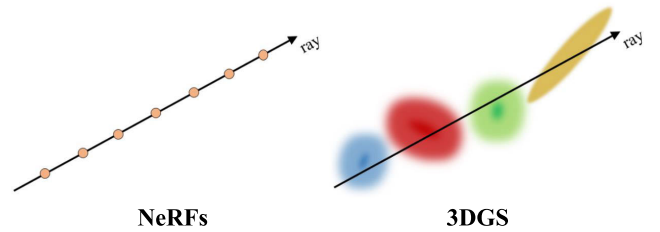


Fig. 7. NeRFs and 3DGS differ conceptually. (Left) NeRFs query an MLP along each ray to synthesize views, whereas (right) 3DGS blends Gaussians along a given ray for representation.

suitable for large-scale environments. This approach comprises a bottom-up hierarchical bundle adjustment and a top-down pose graph optimization. ImMesh [85], leveraging triangular mesh partitioning, enabled online reconstruction of large-scale scenes.

In recent years, significant progress has been made in SLAM methods based on Neural Radiance Fields (NeRFs) [18] and 3D Gaussians Splatting (3DGS) [19], Fig. 6 illustrates the fundamentals of NeRFs and Fig. 7 illustrates the differences between NeRFs and 3DGS. SLAM serves as a pose provider for NeRFs, and NeRFs empower SLAM with the ability to edit scene maps, further enhancing map quality.

iMAP [88] is the first NERF-based SLAM work, capable of dynamically creating implicit 3D models of specific scenes. NICE-SLAM [89] builds upon iMAP, optimizing it with feature grids and a coarse-to-fine approach to address issues of excessive reconstruction and scalability limitations in larger scenes. ESLAM [90] employs Truncated Signed Distance Field (TSDF) as the geometric representation, significantly improving convergence speed and reconstruction quality. GO-SLAM [91] optimizes both global poses and 3D reconstructions. Given that Lidar sensors provide accurate depth information over long distances and under various outdoor conditions, NeRF-LOAM [92], proposed by Junyuan Deng et al., adopts a dynamic voxel embedding memory allocation strategy to ensure efficient utilization of computational resources, supporting its stable application in large-scale outdoor scenes. In dataset tests and real-world experiments, NeRF-LOAM demonstrates superior reconstruction quality compared to other methods. However, it does not operate in real-time. To further improve speed, LONER [93] is the first real-time LSLAM method with an implicit neural representation. Moreover, LONER innovatively proposes a new loss function, significantly accelerating convergence and enhancing reconstruction accuracy.

To further enhance the speed of scene reconstruction, 3DGS proposes rasterizing a set of Gaussian ellipsoids to approximate the appearance of 3D scenes, enabling fast convergence (approximately 30 minutes) and real-time rendering ($> 30FPS$) at 1080P resolution. GS-SLAM [94] introduces an adaptive 3D Gaussian expansion strategy, where new 3D Gaussians are added during the training phase, and unreliable 3D Gaussians with captured depth and low rendering opacity values are removed. For ease of initialization, [95] initializes Gaussians using lidar point clouds and establishes an adaptively-sized voxel grid for the global map. CG-SLAM [96] incorporates an uncertainty map during training, significantly improving reconstruction quality. VastGaussian [97] is the first large-scale, high-quality reconstruction and real-time rendering method based on 3DGS. Its introduced progressive data segmentation strategy allows for independent unit optimization and seamless merging, resulting in a complete scene with sufficient 3D Gaussians. Octree-GS [98] presents an octree-structured 3D Gaussian volume rendering approach that significantly boosts real-time rendering performance in city-scale scenes. Street Gaussians [99] introduces a pose optimization tracking strategy and a 4D spherical harmonics appearance model specifically designed for moving vehicles, enabling the reconstruction of complex and dynamic street environments. These reconstructed data will be utilized in subsequent closed-loop simulation tests for AVs, providing robust support for enhancing the reliability and safety of autonomous driving technologies.

In summary, SLAM methods based on NERFs and 3DGS represent a current research hotspot in the field of AD. However, numerous challenges remain in handling dynamic driving environments and complex street scenes. Additionally, enhancing rendering speed to meet real-time requirements is an urgent issue that needs to be addressed.

c) *Collaborative SLAM*: Compared to individual vehicles, cooperative vehicles exhibit significant advantages in environmental perception and self-localization capabilities. Through communication and collaboration among each other, cooperative vehicles can rapidly cover the entire environment. By fusing the observational data and navigation information from each cooperative vehicle, not only is the accuracy of localization significantly improved, but also the potential errors that may arise during mapping are substantially reduced.

The collaborative SLAM architecture can be categorized into centralized [100], distributed, and distributed-centralized [14], Fig. 8 is a schematic diagram of the different architectures. In a centralized architecture, the base station plays a key role in collecting data from all vehicles and processing it centrally to get the best overall trajectory and map estimates. Conversely, in the distributed architecture, vehicles exchange local information with their neighboring vehicles and rely on distributed optimization protocols among them to independently estimate maps. Compared to centralized architecture, distributed architecture indeed exhibits superior robustness; however, its implementation in practical applications of collaborative SLAM is more complex. This complexity stems primarily from the need for local information exchange between vehicles, coordination of distributed optimization protocols, and the requirement for autonomous decision-making capabilities among vehicles. Finally, the distributed-centralized architecture combines the advantages of both and enhances the stability of the system.

To alleviate the pressure of data transmission and processing in centralized architectures, TES-CVIDS proposed in [101] adopts a transmission strategy based on submaps instead of full images, significantly reducing the load on the system's transmission bandwidth. Simultaneously, the server side employs multithreaded parallel processing to effectively segregate keyframe processing, pose graph optimization, and global map construction tasks, thereby ensuring the efficient real-time performance of the TES-CVIDS system. CCM-SLAM [106] proposed by Shuhuan Wen et al., is an innovative centralized SLAM collaboration and sharing framework that leverages the advantages of edge computing to successfully address positioning and map-building issues in dynamic communication network environments. Furthermore, this framework allows the server to send messages to vehicles at more flexible frequencies based on actual demands, significantly reducing unnecessary time delays caused by fixed transmission frequencies and enhancing the system's response speed and efficiency. Considering the risks of server failure or communication constraints in centralized architectures, researchers are also exploring and optimizing collaborative SLAM systems with distributed architectures. RDC-SLAM [107] achieves maximum likelihood estimation of multiple trajectories through distributed graph optimization methods and designs a communication protocol to simplify deployment and efficiently correlate data among multiple vehicles. Verified through a series of practical experiments, RDC-SLAM demonstrates excellent performance in terms of accuracy, computational efficiency, and data efficiency. To handle the numerous dynamic

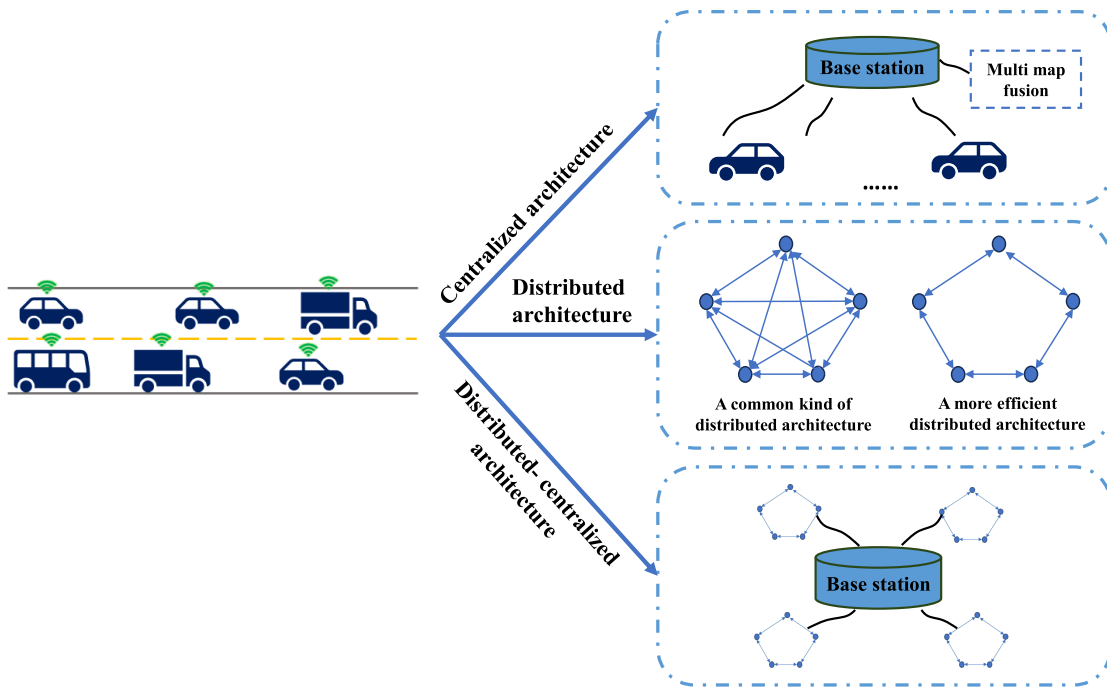


Fig. 8. Three typical architectures for multi-vehicle collaborative SLAM each possess unique characteristics, necessitating the selection of an appropriate architecture based on the specific requirements of the task at hand.

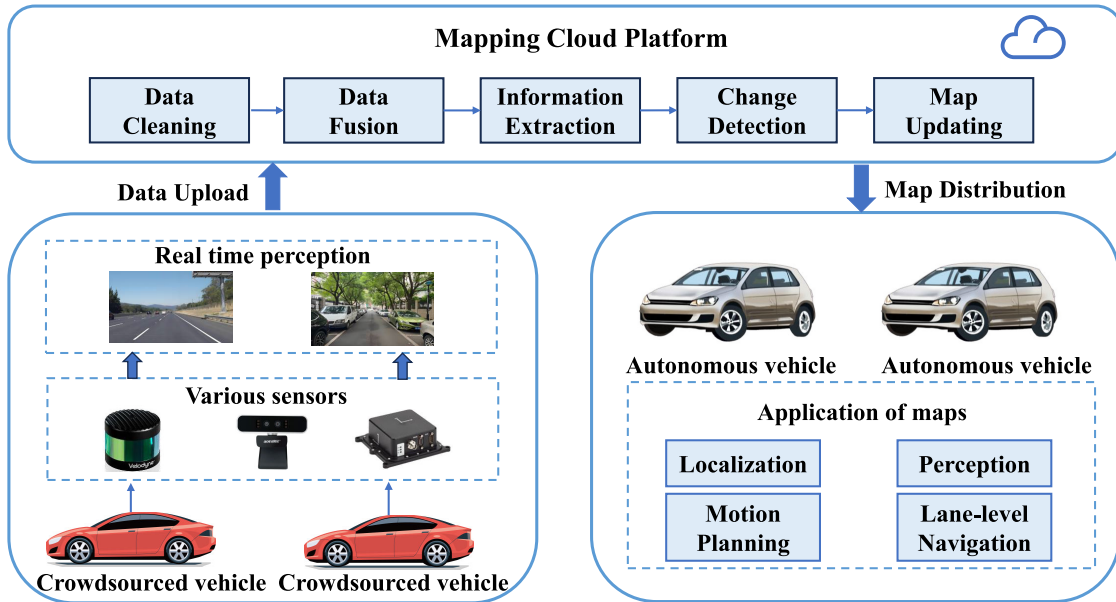


Fig. 9. The overall framework for leveraging crowdsourced information to update HD maps involves crowdsourced vehicles collecting data and uploading it to the cloud, where the data is processed. Subsequently, the processed map is distributed to AVs, facilitating tasks such as localization, perception, planning, and control.

objects present in real-world driving environments, Hao Li et al. proposed a tightly coupled multi-vehicle collaborative SLAMMODT [108] algorithm based on lidar, which performs well in typical and dense dynamic vehicle intersection scenes.

To save hardware costs, reference [109] first proposed the CVIDS, an intensive collaborative SLAM system that does not rely on depth sensors. Leveraging the parallel computing capabilities of GPUs, it significantly enhances

the system's computational efficiency. In [102], a Reconfigurable Holographic Surface (RHS)-assisted SLAM system integrated with federated learning was introduced. This system replaces the expensive phased array antennas traditionally used for millimeter-wave radars with low-cost RHS metasurface antennas, thereby drastically reducing the hardware costs of autonomous driving systems. Additionally, through the federated learning framework, multiple vehicles can collaborate,

TABLE IV

MAP CONSTRUCTION METHODS REGARDING SLAM CRITERIA. “✓”: CRITERION SATISFIED. “×”: CRITERION NOT SATISFIED. “—”: NOT MEASURED.
D: ANTI INTERFERENCE ABILITY FOR DYNAMIC ELEMENTS, L: ANTI INTERFERENCE ABILITY FOR LIGHT CHANGE, S: ANTI INTERFERENCE ABILITY FOR SCENE VARIATION

Method	FPS	ATE(m)	Robustness			System cost	
[71]	43	≤ 0.200 (outdoor)	D:—	L:—	S:✓	Time cost:✓	Hardware cost:✓
[72]	—	≤ 0.880 (outdoor)	D:✓	L:✓	S:✓	Time cost:✓	Hardware cost:✓
[77]	25	≤ 1.830 (outdoor)	D:×	L:—	S:✓	Time cost:✓	Hardware cost:× (Need GPU acceleration)
[79]	18	— (outdoor)	D:✓	L:—	S:—	Time cost:✓	Hardware cost:✓
[82]	—	≤ 0.140 (indoor)	D:×	L:—	S:—	Time cost:—	Hardware cost:× (Large memory usage)
[84]	15	≤ 0.700 (outdoor)	D:×	L:✓	S:✓	Time cost:✓	Hardware cost:✓
[89]	9	≤ 0.100 (indoor)	D:—	L:✓	S:✓	Time cost:×	Hardware cost:✓
[93]	13	— (outdoor)	D:—	L:—	S:✓	Time cost:✓	Hardware cost:✓
[96]	—	≤ 0.080 (indoor)	D:—	L:✓	S:✓	Time cost:✓	Hardware cost:✓
[97]	171	— (outdoor)	D:✓	L:✓	S:✓	Time cost:✓	Hardware cost:× (Need GPU acceleration)
[99]	133	— (outdoor)	D:✓	L:—	S:✓	Time cost:—	Hardware cost:✓
[101]	—	≤ 0.066 (indoor)	D:✓	L:—	S:—	Time cost:—	Hardware cost:—
[102]	17	≤ 2.500 (outdoor)	D:✓	L:—	S:✓	Time cost:✓	Hardware cost:✓
[103]	—	≤ 0.121 (outdoor)	D:✓	L:—	S:—	Time cost:✓	Hardware cost:✓
[104]	21	— (outdoor)	D:✓	L:—	S:✓	Time cost:✓	Hardware cost:✓
[105]	31	≤ 0.240 (outdoor)	D:✓	L:✓	S:✓	Time cost:✓	Hardware cost:✓

share, and fuse sensory data, ultimately enhancing the performance of the SLAM system substantially.

d) Map lightweight and crowdsourced mapping: Given the constraints imposed by the limited onboard computing resources and storage space of AVs, achieving lightweight SLAM mapping becomes particularly crucial. Lightweight SLAM mapping not only reduces the computational complexity of the algorithm, significantly enhancing its operational efficiency, but also drastically decreases the storage requirements for map data. This enables vehicles to store richer map information, effectively alleviating the burden on storage space. Furthermore, the advancement of crowdsourced mapping technology not only forms a beneficial complement to lightweight SLAM mapping but also leverages the advantages of multi-source data and the crowdsourcing model to provide a more efficient, flexible, and widely covered approach for map construction.

In [110], tailored for urban environments, a lightweight feature map based on curbstones and pole-like features is ingeniously designed for lidar-based localization. Simultaneously, a novel, efficient, and precise feature extraction strategy is proposed, realizing a complete pipeline from feature extraction to map construction and finally to localization, demonstrating remarkable practicality and performance. RLD-SLAM [54] designs a robust and lightweight visual-inertial SLAM system utilizing semantic and motion information, enabling it to adapt to various inputs and dynamic scenes. Reference [111] combines feature-based visual odometry with learning-based depth prediction and semantic image segmentation techniques for recognizing and reconstructing semantically relevant environmental structures. This approach achieves comparable reconstruction quality to state-of-the-art voxel-based methods while exhibiting more lightweight advantages in terms of storage and computation. EgoVM [103] introduces a novel end-to-end localization network that enhances localization

accuracy to centimeter-level using lightweight vectorized maps in various challenging urban scenarios. Furthermore, by incorporating lidar geometric features, the localization performance is further improved, compensating for the insufficiency of appearance features in some scenes.

Crowdsourced mapping techniques [112] leverage sensors installed on vehicles to achieve real-time and precise perception of environmental information such as lane markings and traffic signs. The collected data is centrally processed by a cloud-based mapping platform and then transmitted to AVs to assist in navigation, high-precision localization, environmental perception, and path planning. This process is illustrated in Fig. 9. Crowdsourced mapping enables rapid updates of frequently changing map information in driving scenes, ensuring that AVs always have access to the latest and most accurate map data to adapt to complex and dynamic road environments. Reference [112] proposes a crowdsourcing framework that integrates road semantic information collected from multiple vehicles. In [113], researchers introduce an end-to-end crowdsourced mapping framework based on pose graph optimization. Evaluation on a dataset covering over 180 kilometers demonstrates that this framework significantly improves mapping accuracy. Lane information is crucial for the safe and efficient operation of vehicles in the AD domain. Shu et al. [104] propose a lane-level mapping scheme that directly extracts lane-level road information from raw crowdsourced GPS trajectories, enhancing efficiency and accuracy. Reference [114] proposed Crowd-Sourced NeRF, which uses the NeRF model to process large amounts of data acquired by crowdsourced vehicles, and this method solves the key problem of large-scale reconstruction.

To further reduce resource utilization and achieve collaborative computing between AVs and edge servers. Yuru Zhang et al. [105] propose the DATE algorithm, leveraging asynchronous multi-agent reinforcement learning and expected

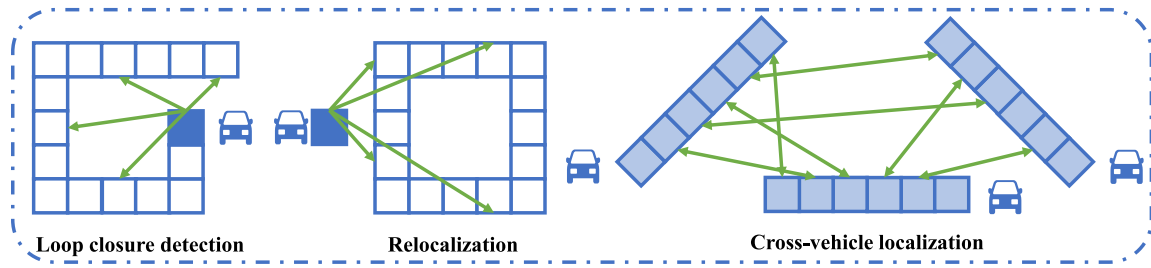


Fig. 10. Global localization techniques employed in AD encompass loop closure detection, relocalization, and cross-vehicle localization, all of which share a common theoretical foundation.

gradient descent, to optimize vehicle offloading and resource reservation across domains, cutting total resource usage by roughly 30% compared to other methods.

A summary of some methods for map construction in large-scale complex scenes is presented in Table IV.

C. Global Localization and Lifelong SLAM

The realization of autonomous navigation is paramount for high-level AVs. In today's era, autonomous driving tasks demand navigation systems capable of long-term operation in large-scale and constantly evolving environments, ultimately aiming for lifelong SLAM [115]. Against this backdrop, the introduction and application of global localization technologies are of utmost importance. In the realm of autonomous driving, global localization primarily encompasses three aspects: relocalization, loop closure detection, and cross-vehicle localization, as depicted in Fig. 10. Relocalization refers to the computation of a relative 6-degree-of-freedom (DOF) transformation of the vehicle's current scan-frame position to a small a priori map of the global coordinate system, which is typically applied when local localization estimation fails. Meanwhile, loop closure detection focuses on identifying locations that the vehicle has previously visited, utilizing this recognition process to correct accumulated drift resulting from prolonged pose estimation. Despite their differing specific objectives and application scenes within AD, both relocalization and loop closure detection rely on similar concepts and theoretical foundations. Initially, the most similar candidates in the prior map to the current scan frame are identified as preliminary references for localization. Subsequently, based on these candidates, a more refined relative pose estimation is performed to achieve accurate global localization.

Cross-vehicle localization involves utilizing incremental SLAM or other advanced mapping techniques to generate multiple online maps from multiple vehicles, which may potentially have partial overlaps. Its objective is to accurately map the position of one vehicle onto the map constructed by another vehicle, enabling global localization across vehicles. Theoretically, the cross-vehicle localization problem is inherently similar to the relocalization problem. Therefore, the relevant technologies are equally applicable to the merging and optimization of offline maps across multiple vehicles.

For AD, the ever-changing environmental conditions under which vehicles operate, including variations in lighting, weather, and seasons, as well as the diverse and complex environments traversed by vehicles, such as open streets,

residential areas, and commercial districts, pose heightened demands on the realization of global localization and lifelong SLAM. These dynamic and diverse environments not only require the system to possess a high degree of robustness and adaptability but also necessitate its ability to continuously learn, update, and maintain map information to adapt to the challenges presented by different environmental types.

a) *Sensor characteristics and multi-sensor fusion:* The effectiveness of global localization hinges on sensor adaptability, characterized by unique attributes listed in Table V. Selecting sensors requires considering factors like field of view (FOV), geometric information richness, and robustness against environmental noise. Vision-based global localization has attracted significant research [116], [117], [118]. It involves two types of cameras: partially-observable and fully-observable. Fully-observable cameras, including pinhole, fisheye, and stereo cameras, are widely used due to their 360-degree panoramic FOV, which ensures viewpoint invariance and reliability in global localization. In contrast, partially-observable cameras, with limited FOVs, are disadvantageous for this task. Compared to vision sensors, lidars capture 3D environmental structures using laser beams, offering stable performance across lighting, weather, and seasons. Recently, FMCW scanning radars have been explored for positioning [119], offering advantages such as lighting insensitivity, robustness under extreme weather (e.g., fog, rain, snow), and 360-degree FOV, making them suitable for challenging outdoor tasks. However, radar devices have lower spatial resolution and generate more noise compared to cameras and lidars.

By carefully evaluating these characteristics, researchers and engineers can select the most appropriate sensors for their specific applications. Moreover, the integration of multiple sensors through multi-sensor fusion can leverage the strengths of each modality, compensating for their individual limitations. Some researchers have proposed multimodal global localization methods, leveraging the unique adaptive characteristics of each sensor modality. There are primarily two approaches: firstly, integrating different sensor modalities to achieve enhanced place recognition [120], thereby improving localization accuracy and robustness; secondly, leveraging the commonalities exhibited by different sensor modalities on the same geometric structures, cross-modal position recognition methods enable efficient and low-cost relocalization on existing maps [121].

TABLE V
CHARACTERISTICS OF THE DIFFERENT SENSORS IN THE GLOBAL LOCALIZATION

Sensors	Global Localization Properties
Partial-observable Camera	Such cameras are widely used in visual global localization tasks, but are sensitive to viewpoint differences, environmental changes, and dynamic objects
Fully-observable Camera	The 360-degree camera provides rich geometric information, which is invariant to viewpoints, but is sensitive to appearance changes and dynamic objects
Lidar	It can provide more accurate 3D environment measurement, but has low resolution and lack of environmental texture input
Radar	The radar has a 360-degree-view, which is insensitive to the appearance and extreme weather, but it has a low resolution and is vulnerable to external noise
Joint Sensors	The combined use of multiple sensors can compensate for the lack of a single sensor

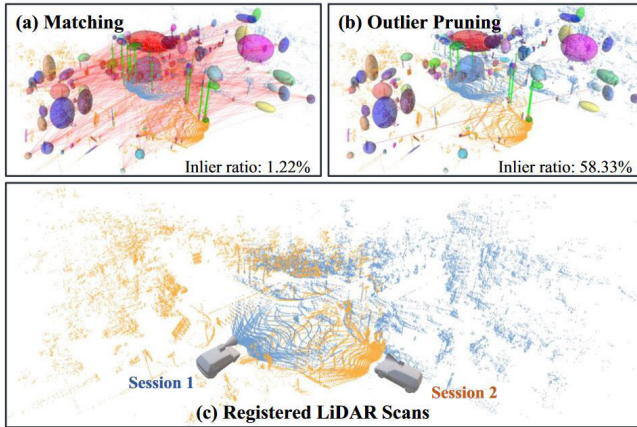


Fig. 11. (a) Front-end Gaussian modeling and correspondence establishment. (b) Back-end outlier pruning increases the inlier ratio of correspondences. (c) Achieving a challenging global registration result at a road intersection, where the two LiDAR point clouds exhibit low overlap and significant view differences [122].

However, the integration of multiple sensing modalities into a cohesive AD system presents both opportunities and challenges. While combining radar, vision systems, and other sensors with SLAM can significantly enhance environmental perception and decision-making capabilities, it also necessitates addressing issues such as data synchronization, spatial alignment, noise reduction, and computational efficiency. Effective multi-sensor fusion requires sophisticated algorithms to harmonize these diverse data streams, ensuring accurate and reliable information for autonomous navigation.

b) Robustness to appearance change: Due to the complexity of vehicle driving environments, the same environment often exhibits varied appearance features [123]. This dynamic change in appearance can not only lead to the failure of recognition in specific areas but also induce perceptual confusion where similar observations arise from different locations. We believe that there are primarily two categories of causes for appearance change. Firstly, variations caused by lighting conditions, weather situations, and seasonal changes significantly impact visual perception, not only increasing the risk of perceptual aliasing but also potentially causing errors in data association. Secondly, structural changes play a crucial role, encompassing various dynamic objects encountered during

navigation, such as moving vehicles or pedestrians, as well as alterations in road geometry, like temporary road diversions due to construction or updates to traffic signs. Additionally, changes in terrain and landforms are also involved, which may stem from natural factors but are more frequently influenced by human activities, particularly construction and urban development.

To effectively address the issue of appearance change, there are currently two primary research approaches. On the one hand, mathematical modeling of locations is employed to extract and enhance those features that remain constant, thereby achieving robust recognition despite changes in appearance. On the other hand, the use of sequential data matching is employed to reduce the impact of appearance change.

Location modeling: Early global localization primarily relied on handcrafted features, as exemplified in [124] and [125]. These classical methods constructed robust environmental descriptions by inducing and summarizing the statistical properties of manually designed features to form a global image representation. In recent years, learning-based features have gradually replaced handcrafted features. Patch-NetVLAD [126] combines the advantages of local and global descriptor methods by deriving patch-level features from NetVLAD residuals. And many variants of NetVLAD continue to emerge, such as TransVLAD [127] and MultiRes-NetVLAD [128], further enhancing the robustness of global features. SRALNet [129] proposed by Peng et al., introduces an interpretable local weighting scheme for reconstructing local features and proposes an attention mechanism that integrates semantic priors with data-driven learning. Recently, with the widespread application of Vision Transformer [130] in the field of visual perception, its potential in visual global localization has attracted significant attention from researchers. R2Former [131] significantly outperforms state-of-the-art methods on benchmark datasets while requiring much less inference time and memory consumption. As shown in Fig. 11, Reference [122] extracts essential geometric primitives such as planes, clusters, and lines (PCL) from the raw point cloud data, transforming them into low-level semantic segments. To represent these segments coherently, each is modeled as a unified Gaussian Ellipsoid Model (GEM), thus facilitating further analysis and processing. While location

TABLE VI

GLOBAL LOCALIZATION METHODS REGARDING SLAM CRITERIA. “✓”: CRITERION SATISFIED. “×”: CRITERION NOT SATISFIED. “—”: NOT MEASURED. D: ANTI INTERFERENCE ABILITY FOR DYNAMIC ELEMENTS, L: ANTI INTERFERENCE ABILITY FOR LIGHT CHANGE, S: ANTI INTERFERENCE ABILITY FOR SCENE VARIATION

Method	FPS	ATE(m)	Robustness			System cost	
[127]	23	—(outdoor)	D:✓	L:✓	S:✓	Time cost:✓	Hardware cost:—
[129]	—	≤0.212(outdoor)	D:✓	L:—	S:✓	Time cost:—	Hardware cost:✓
[131]	16	≤0.810(outdoor)	D:✓	L:—	S:✓	Time cost:×	Hardware cost:× (Need GPU acceleration)
[132]	10	≤1.120(outdoor)	D:✓	L:—	S:—	Time cost:✓	Hardware cost:× (Large memory usage)
[133]	27	≤0.631(outdoor)	D:—	L:—	S:✓	Time cost:—	Hardware cost:✓
[134]	20	≤0.176(outdoor)	D:✓	L:✓	S:✓	Time cost:✓	Hardware cost:✓
[135]	17	—(outdoor)	D:✓	L:✓	S:✓	Time cost:✓	Hardware cost:✓
[136]	18	≤1.710(outdoor)	D:×	L:—	S:✓	Time cost:✓	Hardware cost:✓
[137]	15	≤1.500(outdoor)	D:✓	L:—	S:✓	Time cost:✓	Hardware cost:✓
[138]	22	≤1.800(outdoor)	D:—	L:—	S:✓	Time cost:✓	Hardware cost:✓
[139]	—	≤3.360(outdoor)	D:—	L:—	S:✓	Time cost:—	Hardware cost:✓
[140]	—	—(outdoor)	D:✓	L:✓	S:—	Time cost:✓	Hardware cost:✓
[141]	25	≤2.120(outdoor)	D:✓	L:✓	S:✓	Time cost:✓	Hardware cost:✓
[142]	—	≤4.695(outdoor)	D:✓	L:—	S:—	Time cost:—	Hardware cost:✓
[143]	—	—(outdoor)	D:✓	L:✓	S:✓	Time cost:✓	Hardware cost:—

hybrid approaches. AutoMerge [143], as a representative of hybrid methods, combines point-based geometric feature extraction with projection-based appearance feature extraction. This comprehensive approach effectively tackles the challenges arising from translational and rotational variations, offering a more resilient and efficient framework for global localization across diverse viewpoints.

Table VI provides an assessment of diverse SLAM methods focused on achieving global localization.

III. ORGANIZATION OF RELATED DATASETS

Currently, the research on SLAM algorithms remains heavily reliant on open-source datasets, which serve as the cornerstone for evaluation and comparison, ensuring fairness and comparability in research endeavors. By conducting tests within a unified dataset framework, researchers worldwide can transcend geographical boundaries to conduct horizontal comparative analyses of their algorithm’s performance, thereby fostering continuous advancements in SLAM technology. In light of this, we have collected and compiled 20 datasets encompassing complex AD scenes, as shown in Table VII. These datasets are designed to comprehensively address the diverse challenges faced in AD, including but not limited to complex urban traffic, rugged off-road terrain, extreme weather conditions, and more. Our primary goal is to furnish robust data support that enables the achievement of high-level AD capabilities.

IV. CHALLENGES AND OPPORTUNITIES

In this section, we will highlight some of the challenges and opportunities faced by SLAM in complex autonomous driving scenes, and summarize current as well as potential future research directions. An outline of the relevant content is presented in Fig. 13, aiming to provide valuable reference and inspiration for researchers in the field of AD.

A. Long Term SLAM in Dynamic Scenes

When discussing SLAM systems, the impact of dynamic objects cannot be overlooked, with the crux lying in how to effectively eliminate and track these objects. For AD, achieving precise localization and mapping in environments dense with dynamic objects is of paramount importance. To this end, we must continue to delve into advanced technologies such as incremental map updating and dynamic environment modeling, ensuring that AVs can operate stably and accurately in complex and ever-changing dynamic environments.

Despite the progress made by existing dynamic object modeling algorithms, their performance remains inadequate when dealing with non-rigid objects. We believe that it is necessary to conduct deep optimization of SLAM systems to enhance their long-term stable operation performance in dynamic environments, particularly in terms of modeling capabilities for non-rigid dynamic objects.

B. SLAM in Complex and Unstructured Outdoor Environments

As the technology of AD progresses, the operational environments of vehicles are no longer limited to traditional structured urban scenes. They will expand into more complex and variable unstructured environments, such as dense forests, vast deserts, and rugged wildernesses. AVs face significant challenges in unstructured environments, as these often present difficulties in extracting effective feature information, significantly impacting the vehicles’ positioning accuracy. More crucially, the absence of prominent landmark features severely affects the accuracy of global localization, making it challenging to effectively correct cumulative errors. Therefore, we believe that researching SLAM methods in unstructured environments is not only necessary but also a vital step towards achieving higher levels of AD.

TABLE VII
20 COMPLEX AUTONOMOUS DRIVING SCENES DATASETS FOR SLAM

Datasets	Year	Weather	Time of day	Scenes conditions	Link	Volume	R & S
WHUVID [147]	2022	sunny, cloudy	day, afternoon, night	complex urban, dynamic objects	https://github.com/chentianyangWHU/WHUVID	82km	real
NCLT [148]	2016	variety	from morning to dusk	indoor and outdoor, seasonal variation	http://robots.engin.umich.edu/nclt	147.4km	real
CARLA-Loc [149]	2023	heavy rain, haze, snow	day, night	dynamic scenes, reflective objects	https://yuhang1008.github.io/CARLA-Loc_page/	38.3km	synthetic
KAIST [150]	2019	sunny, cloudy	daytime	complex urban, large scene	http://irap.kaist.ac.kr/dataset	40 sequences	real
RADIATE [151]	2021	variety	day, night	urban, countryside, highway	http://pro.hw.ac.uk/radiate/	44K images	real
4Seasons [152]	2022	variety	variety	seasonal variation, urban, countryside, highway	https://www.4seasons-dataset.com	350km	real
MulRan [153]	2020	sunny, cloudy	unknow	few structural features	https://sites.google.com/view/mulran-pr	41.2km	real
CCSAD [154]	2015	sunny	daytime	urban, poor road conditions	camaron.cimat.mx/Personal/jbhayet/ccsad-dataset	500GB	real
NVSEC [155]	2018	sunny, cloudy	day, night	indoor and outdoor	https://daniilidisgroup.github.io/mvsec	28 km	real
SubT-MRS [156]	2024	variety	variety	various challenging scenes	https://superodometry.com/datasets	300km	real
UrbanLoco [157]	2019	sunny, cloudy	day, night	complex urban, dynamic objects	https://github.com/weisongwen/UrbanLoco	40km	real
DSEC [158]	2021	sunny	variety	light variation, unstructured environment	http://rpg.ifi.uzh.ch/dsec.html	4TB	real
ECMD [159]	2024	sunny	day, night	high speed, repetitive scenes, dynamic objects	https://arclab-hku.github.io/ecmd/	200km	real
OpenMPD [160]	2022	variety	daytime	urban crowded environment and unstructured roads	https://arclab-hku.github.io/ecmd/	1.5K frames	real
DIDLm [161]	2024	rain and snow	day, night	rugged roads, indoor park lot	https://github.com/GongWeiSheng/DIDLm	250GB	real
UTBM [162]	2020	sunny, cloudy, snow	day, dusk, night	highly dynamic environment, long-term autonomy	https://epan-utbm.github.io/	63.4km	real
M2DGR [163]	2021	unknow	From morning to dusk	indoor and outdoor, multiple scenes	https://github.com/SJTU-ViSYS/M2DGR	36 sequences	real
OORD [164]	2024	Poor weather	variety	rough cross-country road, unstructured	https://oxford-robotics-institute.github.io/oord-dataset/	154km	real
TartanAir [165]	2020	variety	variety	multiple challenging scenes	https://theairlab.org/tartanair-dataset/	4TB	synthetic
SUPS [166]	2022	unknow	unknow	Underground Parking	https://github.com/jarvishou829/SUPS	5K frames	synthetic

R: Real

S: Synthetic

C. Algorithm Deployment on Low Computing Power Platforms

For AVs, executing the SLAM algorithm module is merely one part of their complex tasks, followed by the operation of decision-making planning and motion control modules. This series of operations poses high demands on the computational resources of vehicle embedded platforms. In reality, while many advanced SLAM algorithms demonstrate exceptional performance, they often require significant computational resources, which to some extent limits their ability to meet real-time requirements in practical applications. Given the stringent real-time requirements of AD systems, lightweight design of SLAM algorithms to reduce their computational complexity and enhance execution efficiency has become an important research direction.

D. Generation of Test Data Related to Autonomous Driving

In the development process of AVs, ensuring their safety during real-world operation is paramount, which relies on a

series of thorough and precise tests. These tests often necessitate vast amounts of data to simulate and cover a wide range of possible driving scenes. With the advancement of rendering technologies such as Neural Radiance Fields (NeRFs) and 3D Gaussian Splatting (3DGS) processes, it becomes feasible to accurately simulate and reproduce scenes from collected data, and further generate entirely new driving environments, thereby significantly enriching our test datasets.

As rendering quality continues to improve and computational demands decrease, these methods have demonstrated the potential to replace some real-world testing scenes. They not only provide realistic simulated environments but also effectively reduce the physical burden during testing, opening up new possibilities for the development and testing of AVs.

E. Integration With Large Language Models

With the development of deep learning, Large Language Models (LLMs) have also found applications in the SLAM domain, elevating SLAM perception to a new level of natural

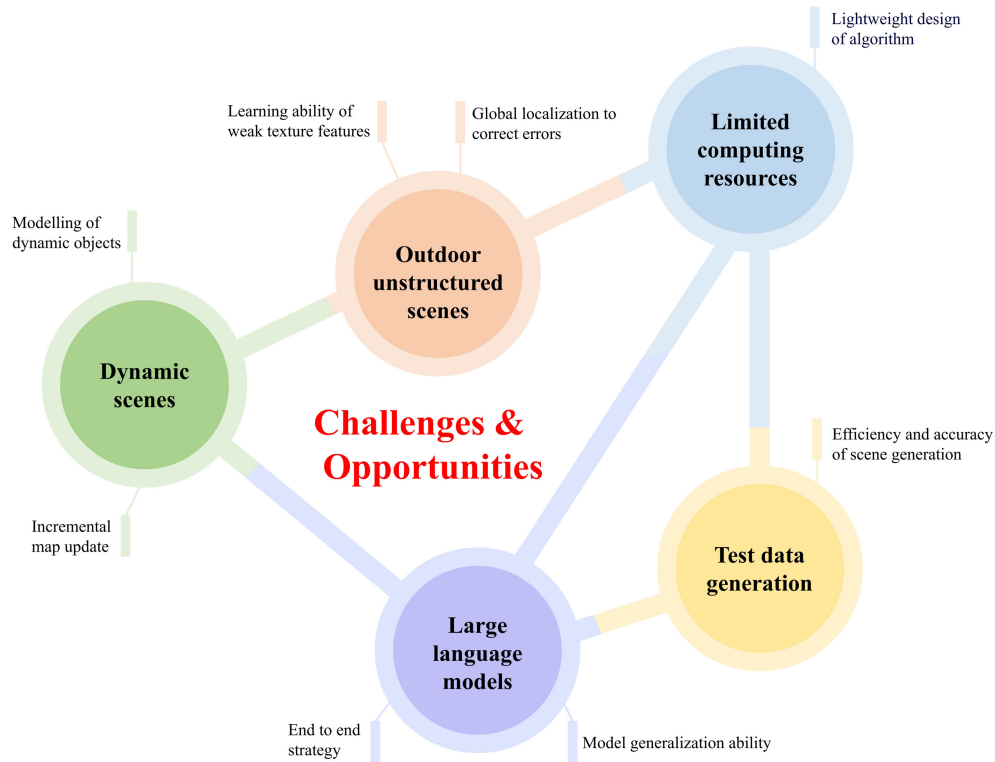


Fig. 13. Illustration of potential research directions.

language understanding. By learning normal and abnormal patterns, LLMs can detect and identify anomalies in the environment, enhancing the robustness and reliability of SLAM systems and providing a solid foundation for the safe operation of AVs. We believe that further research in this area is warranted. On one hand, leveraging large models can significantly improve environmental understanding capabilities, enabling SLAM systems to perceive the surrounding environment with greater precision. On the other hand, large models can predict and analyze environmental changes, providing proactive guidance to the system. Additionally, large models can efficiently process multi-sensor data, including cameras, LiDARs, and infrared sensors, achieving comprehensive information fusion.

V. CONCLUSION

This paper provides a comprehensive review of the current research status of SLAM methods in complex AD scenes, delving into the challenges faced, advancements achieved, and their practical value in applications. Our analysis revolves around three core perspectives: Firstly, the advantages and limitations of the current techniques are discussed for the accurate removal and effective tracking of dynamic objects. Secondly, we analyze the technical challenges and solutions in constructing large-scale maps. Lastly, we delve into the accuracy and robustness of global localization techniques. Furthermore, to facilitate further research advancements, we compile a series of relevant datasets for complex AD scenes, aiming to provide researchers with a diverse range of experimental environments and testing resources that positively impact algorithm research and testing. Based on an in-depth analysis of current

research, we discuss potential future research directions and offer constructive suggestions. We hope this article inspires more innovative research, propels the continuous progress of SLAM in the AD field, and lays a solid foundation for the widespread application of AD.

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